

Logic Operations

37) ANA B

38) ANA M

39) ANI 25H

40) ORA B

41) ORA M

42) ORI 25H

43) XRA B

44) XRA M

45) XRI 25H

46) RLC

47) RRC

48) RAL

49) RAR

50) CMP B

51) CPI 25H

Implied instructions
(Works on Acc.)

B =
A =

1) ANA R
 $\Phi: A \wedge B$
 $A \leftarrow A \wedge B$

AND

0	0	0
0	1	0
1	0	0
1	1	1

2) ANA M

3) ANI 25H

OR

0	0
0	1
1	0
1	1

4) ORA R
 $A \leftarrow A \vee R$

5) ORA M

6) ORI 25H

XOR

0	0
0	1
1	0
1	1

7) XRA R
 $A \leftarrow A \oplus R$

8) XRA M

Clear LN

A = 00 11 010

Set LN

A = 00 11 010 1

Complement

A = 00 11 010



1) ANA R
 2) ANA B
 $\leftarrow A \wedge B$

ANA M

3) ANI 25H

4) ORA R
 $\leftarrow A \vee R$

M

25H

R

$\leftarrow A \oplus R$

M

25H

AND

00	0
01	0
10	0
11	1

ANI 0FH

A = 0011 0000

Clear LN

A = 0011	0101 = 35
1111	0000
A = 0011	0000

OR

00	0
01	1
10	1
11	1

ORI 0FH

A = 0011	0101
0000	1111
A = 0011	1111

Set LN

A = 0011	0101
0000	1111
A = 0011	1111

XOR

00	0
01	1
10	1
11	0

XRI 0FH

A = 0011	0101
0000	1111
A = 0011	1010

Complement LN

10) CMP

A > B

A = B

A < B

11) CMP

12) CPI

13) STC

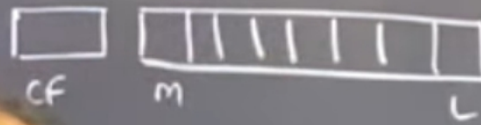
14) CM

15) CM

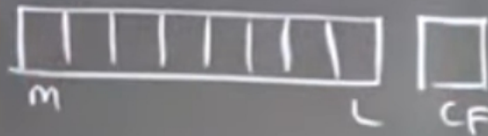
10) CMP B
 $A - B$

	CF	ZF
✓ $A > B$	0	0
✓ $A = B$	0	1
✓ $A < B$	1	0

11) RLC -A-



RC -A-



11) CMP m
 A - m

12) CPI 25H
 $A - 25$

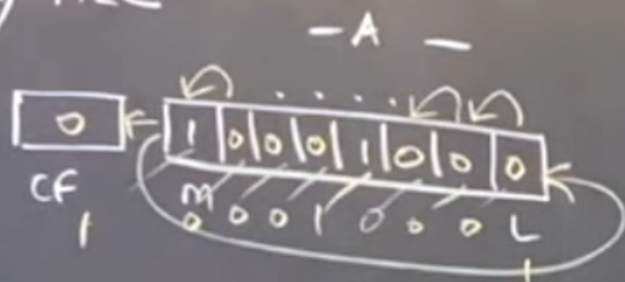
13) STC CF = 1

14) CMC $CF = \overline{CF}$

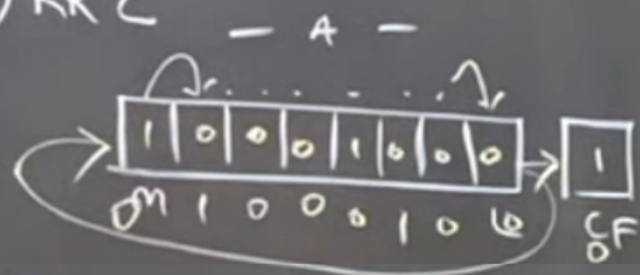
15) CMA $A = \overline{A}$

3

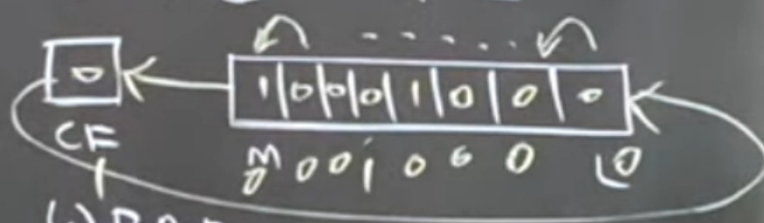
1) RLC



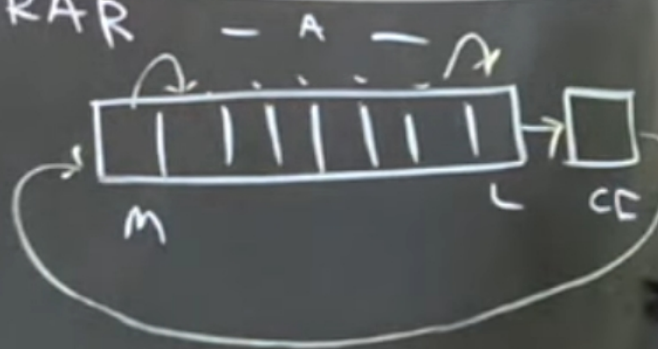
2) RRC



3) RAL



4) RAR



LOGIC GROUP

Special Note:

In the explanation of every instruction, I have mentioned its machine cycles and T-states. You will understand this part once you watch the two videos of timing diagrams

1) ANA R

Logically AND the contents of the specified register with accumulator, store result in accumulator.

Eg: **ANA B** ; A ← A AND B

Addr. Mode	Flags Affected	Cycles	T-States
Register	ALL	1	4

2) ANA M

Logically AND the contents of the memory location pointed by HL pair, with the accumulator.

Eg: **ANA M** ; A ← A AND M

Addr. Mode	Flags Affected	Cycles	T-States
Indirect	ALL	2	7

3) ANI 8-bit data

Logically AND the immediate 8-bit data, with the accumulator.

Eg: **ANA 25** ; A ← A AND 25

Addr. Mode	Flags Affected	Cycles	T-States
Immediate	ALL	2	7

Special Note: Use of AND operation

To "Clear any bit", we must "AND" that bit with "0" and the remaining bits with "1".

Eg: ANI F0H will Clear the Lower Nibble of A while the Higher Nibble will remain the same.

Similarly we have the other logic instructions as follows:

- 4) **ORA R**
- 5) **ORA M**
- 6) **ORI 8-bit data**

Special Note: Use of OR operation

To "Set any bit", we must "OR" that bit with "1" and the remaining bits with "0".

Eg: ORI 0FH will Set the Lower Nibble of A while the Higher Nibble will remain the same.

- 7) **XRA R**
- 8) **XRA M**
- 9) **XRI 8-bit data**

Special Note: Use of XOR operation

To "Complement any bit", we must "XOR" that bit with "1" and the remaining bits with "0".

Eg: XRI 0FH will Complement the Lower Nibble while the Higher Nibble will remain the same.

10) CMP R

Compares the contents of register R and accumulator.

Comparison essentially is subtraction. Hence, this instruction performs $A - R$.

It is very important to **remember** that the **result** of this comparison is **NOT stored in accumulator**, only the Flags are affected. ☺ In case of doubts, contact Bharat Sir: - 98204 08217.

Eg: CMP B ; Compares A and B i.e. $A - B$ (and not $B - A$)

We decide which one of the two is greater by checking the flags affected as follows:

Conclusion	Zero Flag 'Z'	Carry Flag 'Cy'
$A > B$	0	0
$A = B$	1	0
$A < B$	0	1

Addr. Mode	Flags Affected	Cycles	T-States
Register	ALL	1	4

Similarly we have the other comparison instructions as follows:

- 11) **CMP M**
- 12) **CPI 8-bit data**

13) STC

Sets the carry flag.

$Cy \leftarrow 1$.

Addr. Mode	Flags Affected	Cycles	T-States
Implied	Only Carry	1	4

14) CMC

Complements the carry flag.

$Cy \leftarrow Cy$.

Addr. Mode	Flags Affected	Cycles	T-States
Implied	Only Carry	1	4

15) CMA

Complements the accumulator.

$A \leftarrow 1$'s complement of A.

Addr. Mode	Flags Affected	Cycles	T-States
Implied	None	1	4

Special Note: Use of CMA operation

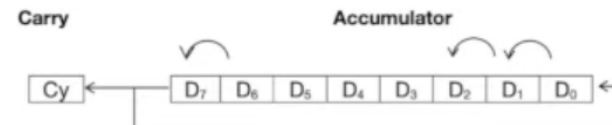
It acts as a NOT gate. AND followed by NOT gives a NAND.

Hence using CMA we can derive NAND, NOR and XNOR operations.

16) RLC

The Contents of accumulator are rotated left by 1.

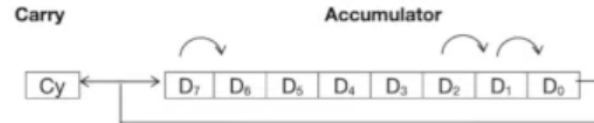
The MSB goes to the Carry AND the LSB.



Addr. Mode	Flags Affected	Cycles	T-States
Implied	Carry	1	4

17) RRC

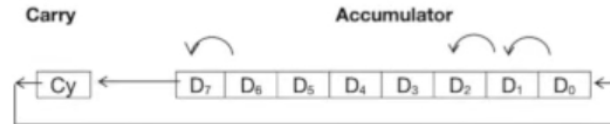
The Contents of accumulator are rotated right by 1.
The LSB goes to the Carry AND the MSB.



Addr. Mode	Flags Affected	Cycles	T-States
Implied	Carry	1	4

18) RAL

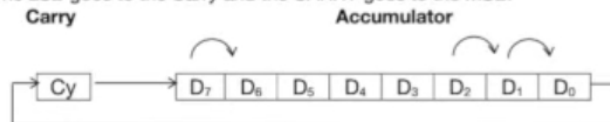
The Contents of accumulator are rotated left by 1.
The MSB goes to the Carry and THE CARRY goes to LSB.



Addr. Mode	Flags Affected	Cycles	T-States
Implied	Carry	1	4

19) RAR

The Contents of accumulator are rotated right by 1.
The LSB goes to the Carry and the CARRY goes to the MSB.



Addr. Mode	Flags Affected	Cycles	T-States
Implied	Carry	1	4